

WHAT IS AUTISM?

Chapter One

Autism is a life-long developmental disability that prevents people from understanding what they see, hear, and otherwise sense. This results in severe problems with social relationships, communication, and behaviour.

The *Diagnostic and Statistical Manual of Mental Disorders*, DSM-IV (American Psychiatric Association, 1994) defines autism as a pervasive developmental disorder characterized by:

- impairments in communication and social interaction, and
- restricted, repetitive, and stereotypic patterns of behaviour, interests, and activities.

It is a complex neurological disorder that affects the functioning of the brain.

Autism symptoms can be present in a variety of combinations and may accompany other disabilities. Some people with autism have normal levels of intelligence, while most people with autism have some level of intellectual disability, ranging from mild to severe. This range is often referred to as high-functioning autism to low-functioning autism.

There may be a range of difficulties in expressive and receptive language and communication. It is estimated that up to 50% of people with autism do not develop functional speech. For those who do, speech may have unusual qualities and have limited communicative functions.

All people with autism have difficulties with social interaction and behaviour, but the extent and type of difficulty may vary. Some individuals may be very withdrawn, while others may be overly active and approach people in peculiar ways. They have problems with inattention and resistance to change. They often respond to sensory stimuli in an atypical manner and may exhibit odd behaviours such as hand flapping, spinning, or rocking. They may also demonstrate unusual uses of objects and attachments to objects.

Although people with autism share some common features, no two individuals are the same. In addition, the pattern and extent of difficulties may change with development. The common characteristics help us to understand general needs associated with autism, but it is important to combine this information with knowledge of the specific interests, abilities, and personality of each student.



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Prevalence of autism

The generally accepted prevalence rate for autism has been between four and five in every 10,000 births. However, some recent estimates suggest a rate of about 10 in 10,000 or 0.1% of children or more, when a broader spectrum of disorders is included. There is a higher prevalence among males. The ratio varies depending on the definition, but studies reveal a male-to-female ratio of between 3:1 and 4:1.

Causes of autism

The cause or combination of causes of autism is not fully known. There is growing evidence that autism is a genetic condition, and that there are likely several different genes involved. The mode of genetic transmission appears complex, and scientists are focusing their work on finding which genes may be involved and how these genes are affected. So far, it appears that for at least a significant subgroup of persons with autism, there is a genetic susceptibility which differs across families (that is, different genes may be responsible in different families).

There is also evidence that there may be a higher prevalence among children with autism of problems very early in the mother's pregnancy, at birth, or even after birth than for children who do not have autism. Early life events and environmental factors may interact significantly with genetic susceptibility in the child.

Recently, various types of investigations, including imaging studies, electro-encephalographic studies, tissue studies on autopsy material, and neuro-chemical studies, have provided further evidence of a biological basis for autism. The brains of individuals with autism appear to have some structural and functional differences from the brains of other people. Anomalies in the brain stem and cranial nerves have been found. Ongoing research may one day pinpoint the exact genes and other conditions that combine to cause autism.

Diagnosis of autism

The diagnosis of autism is made by a physician or clinical psychologist with expertise in the area of autism. Assessment and diagnosis ideally should involve a multidisciplinary team that includes a pediatrician or psychiatrist, a psychologist, and a speech and language pathologist. The psychologist often administers assessments to gather information about developmental level and behaviour, and the speech and language pathologist assesses speech, language, and communicative behaviours. A medical assessment is conducted to rule out other possible causes for the symptoms, as many of the characteristics associated with autism are also present in other disorders. A medical and developmental history is taken through discussion with the parents. This information is combined with the other assessments to provide the overall picture and to rule out other contributing factors.

Parents who are seeking additional information regarding assessment and diagnosis should contact health professionals in their community.

Professionals diagnose autism through the presence or absence of certain behaviours, characteristic symptoms, and developmental delays. The criteria for autism are outlined in the *DSM-IV* and the *International Classification of Diseases* (World Health Organization, 1993).

The *DSM-IV*, which is most commonly used in North America, classifies autism as a disorder within a broader group of pervasive developmental disorders (PDD). PDD is an umbrella term for disorders that involve impairments in reciprocal social interaction skills, communication skills, and the presence of stereotypical behaviours, interests, and activities. The conditions classified as PDD in the *DSM-IV* are:

- Autism
- Childhood Disintegrative Disorder (CDD)
- Rett's Disorder
- Asperger's Disorder
- Pervasive Developmental Disorder Not Otherwise Specified (PDD-NOS)

Some of these diagnostic terms appear to be used interchangeably within the literature and by practitioners. The term *Autism Spectrum Disorders* is sometimes used to refer to autism and other conditions included within the PDD classification. PDD is sometimes used to refer

to all conditions within the category of PDD, and at other times it has been used to refer to PDD-NOS. This confusion of terminology can be a problem when people from different disciplines are working together to support students.

The *DSM-IV* criteria for autism are included below.

Criteria for autistic disorder in the DSM-IV (299.00)

A.. A total of at least six items from (1), (2), and (3), with at least two from (1), and one from (2) and (3):

- (1) Qualitative impairment in social interaction, as manifested by at least two of the following:
 - (a) Marked impairment in the use of multiple non-verbal behaviours such as eye-to-eye gaze, facial expression, body postures, and gestures to regulate social interaction
 - (b) Failure to develop peer relationships appropriate to developmental level
 - (c) Markedly impaired expression of pleasure in other people's happiness.
- (2) Qualitative impairments in communication as manifested by at least one of the following:
 - (a) Delay in, or total lack of, the development of spoken language (not accompanied by an attempt to compensate through alternative modes of communication such as gestures or mime)
 - (b) In individuals with adequate speech, marked impairment in the ability to initiate or sustain a conversation with others
 - (c) Stereotyped and repetitive use of language or idiosyncratic language
 - (d) Lack of varied spontaneous make-believe play or social imitative play appropriate to developmental level.
- (3) Restricted repetitive and stereotyped patterns of behaviour, interests, and activities, as manifested by as least one of the following:
 - (a) Encompassing preoccupation with one or more stereotyped and restricted patterns of interest that is abnormal either in intensity or focus
 - (b) Apparently compulsive adherence to specific nonfunctional routines or rituals

- (c) Stereotyped and repetitive motor mannerisms (e.g., hand or finger flapping or twisting, or complex whole-body movements)
 - (d) Persistent preoccupation with parts of objects.
- B. Delays or abnormal functioning in at least one of the following areas, with onset prior to age three years:
- (1) social interaction,
 - (2) language as used in social communication, or
 - (3) symbolic or imaginative play.
- C. Not better accounted for by Rett's Disorder or Childhood Disintegrative Disorder.

Other Pervasive Developmental Disorders

All of the disorders within the PDD classification have some common features, and children with these disorders may benefit from similar instructional strategies. However, there are differences in some areas, such as the number of symptoms, age of onset, and developmental pattern.

In British Columbia, students who have developmental disorders but are not diagnosed as autistic may be identified for special education funding purposes in the category that best reflects their medical condition and the type and intensity of educational interventions required. In some cases, the appropriate category may be "Students With Intellectual Disabilities." In others, it may be "Severe Behaviour Disorders," "Moderate Behaviour Disorders," or "Physical Disabilities/Chronic Health Impairments." Some students may meet the ministry criteria for "Severe Learning Disabilities."

Asperger's syndrome

Asperger's syndrome shares many of the features of autism. People with Asperger's syndrome have disabilities in the areas of social interaction and stereotypical behaviour patterns. For this reason, Chapter 6 of this resource guide contains specific information about the characteristics of students with Asperger's syndrome and suggestions for classroom strategies.

The main differences between students with autism and those with Asperger's syndrome is that children with AS do not have clinically



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For more information on special education categories in B.C., see the Ministry of Education document *Special Education Services: A Manual of Policies, Procedures and Guidelines*, 1995.

significant delays in early language development or significant delays in cognitive development. They usually do not have the same degree of difficulty as those with autism in the development of age-appropriate self-help skills, adaptive behaviour, and curiosity about the environment in childhood.

The *DSM-IV* uses the term *Asperger's Disorder*. This resource guide uses the term Asperger's syndrome, which is consistent with current literature in the area.

See Chapter 6, *Teaching Students With Asperger's Syndrome*, for more information.

CHARACTERISTICS ASSOCIATED WITH AUTISM

Chapter Two

Although every person with autism is unique, some characteristics are considered to be particularly important in the diagnosis of autism. These fall into four major categories:

- communication characteristics
- social interaction characteristics
- unusual behaviour characteristics
- learning characteristics

Other characteristics of behaviour and learning of students with autism can be categorized as:

- unusual patterns of attention
- unusual responses to sensory stimuli
- anxiety

This chapter addresses these seven categories and outlines instructional implications for each. Later chapters offer more detailed suggestions for teaching students with autism.

Communication

All people with autism experience language and communication difficulties, although there are considerable differences in language ability among individuals. Some are non-verbal while others may have extensive language with deficits in the area of pragmatics (the social use of language). People with autism may seem caught up in a private world in which communication is unimportant. This is not an intentional action but rather an inability to communicate.

Language difficulties that may be present include:

- difficulties with non-verbal communication:
 - inappropriate facial expressions
 - unusual use of gestures
 - lack of eye contact
 - strange body postures



Indiana Resource Centre
for Autism, *Autism
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for children with
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Lindblad, T., "Language
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"The student may be using echolalic utterances to rehearse what is heard in order to process the information, or as a strategy for self-regulation."

B. Prizant & J. Duchan, "The functions of immediate echolalia in autistic children," *Journal of Speech and Hearing Disorders*, 46, 1981



Quill, K. A., "Visually cued instruction for children with autism and pervasive developmental disorders," *Focus on Autistic Behaviour*, 110 (3), 1995.

- lack of mutual or shared focus of attention
- delay in or lack of expressive language skills
- significant differences in oral language, for those who do develop language:
 - odd pitch or intonation
 - faster or slower rate than normal
 - unusual rhythm, or stress
 - monotone or lilting voice quality
- repetitive and idiosyncratic speech patterns
- echolalic speech, immediate or delayed literal repetition of the speech of others:
 - appears to be non-meaningful, but may indicate an attempt to communicate
 - indicates the ability to produce speech and to imitate
 - may serve a communication or cognitive purpose for the student
- restricted vocabulary:
 - dominated by nouns
 - often confined to requests or rejections to regulate one's physical environment
 - limited in social functions
- tendency to persevere on a topic—that is, to continually discuss one topic and have difficulty changing topics
- difficulty with pragmatics of conversation:
 - problems initiating the communication
 - difficulty using unwritten rules
 - inability to maintain conversation on a topic
 - inappropriate interrupting
 - inflexibility in style of conversation, stereotypic style of speaking

People with autism often have difficulty in comprehending verbal information, following long verbal instructions, and remembering a sequence of instructions. The comprehension of language may be context-specific. The extent of difficulty will vary among individuals, but even those who have normal intelligence, usually referred to as high-

functioning, may have difficulty with comprehension of verbal information.

Implications for instruction

Effective programs for students with autism and other pervasive developmental disorders include comprehensive communication assessment and intervention. This typically involves assessment by a speech and language pathologist as well as informal observation and classroom-based evaluation. The assessment serves as the basis for the identification of goals, objectives, and strategies for facilitating development of receptive language and expressive skills, particularly with pragmatic skills. Instruction should emphasize paying attention, imitating, comprehending, and using language in play and social interaction. Communication goals should emphasize the functional use of language and communication in various settings.

For more information on teaching communication skills, see Chapter 4, *Strategies for Communication Development*.

Social interaction

Students with autism demonstrate qualitative differences in social interaction and often have difficulty establishing relationships. They may have limited social interactions or a rigid way of interacting with others. The difficulties they have with social communication should not be seen as a lack of interest or unwillingness to interact with others; this lack of effective communication may result from an inability to distill social information from the social interaction and use appropriate communication skills to respond.

Understanding social situations typically requires language processing and non-verbal communication, which are often areas of deficit for people with autism. They may not notice important social cues, and may miss necessary information. People with autism typically have an impairment in the use of non-verbal behaviours and gestures to regulate social interaction, and they may have difficulty reading the non-verbal behaviour of others.

People with autism have significant difficulty with any interaction that requires knowledge of other people and what they think or know. It has been theorized that people with autism have a social cognitive deficit in this area. Baron-Cohen has described this as the “theory of mind”:

"One must separate the variables of social interaction problems from emotions. People with autism desire emotional contact with other people but they are stymied by complex social interaction."

Temple Grandin, *Thinking in Pictures and Other Reports From My Life With Autism*, 1995, p. 44



For more information on teaching social communication, see Quill, K. A., *Teaching Children with Autism: Strategies to Enhance Communication and Socialization*, 1995.



Sigman, M., et al.,
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*Handbook of Autism and
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Autism to Mind Read: A
Practical Guide*, 1999.



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and classification,"
*Journal of Autism and
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Disorders*, 9, 1979.

people with autism are not able to understand the perspective of others, or even to understand that other people have a perspective that could be different from their own. They may also have difficulty understanding their own—and particularly other people's—beliefs, desires, intentions, knowledge, and perceptions. Students with autism often have problems understanding the connection between mental states and actions. For example, children with autism may not be able to understand that another child is sad—even if that child is crying—because they are not themselves sad.

Teachers may better understand the thinking and behaviour of their students with autism if they realize that these students may not be able to grasp the fact that other people have their own perceptions and viewpoints. Students with autism demonstrate these difficulties in a variety of observable ways. They have a tendency to play with toys and objects in unusual and stereotypical ways. Some may engage in excessive or inappropriate laughing or giggling. Play that does occur often lacks the imaginative qualities of social play. Some children with autism may play near others, but do not share and take turns, while others may withdraw entirely from social situations.

The quality and quantity of social interaction occurs on a continuum. Social interaction can be classified into three subtypes along this continuum:

- aloof—those who show no observable interest or concern in interacting with other people except for those needed to satisfy basic personal needs; they may become agitated when in close proximity to others and may reject unsolicited physical or social contact
- passive—those who do not initiate social approaches, but will accept initiations from others
- active but odd—those who will approach for social interaction but do so in an unusual and often inappropriate fashion

It should be noted that people with autism do not necessarily fall into one distinct spot on the continuum.

Implications for instruction

Social skill development is essential for students with autism, and it is an important component in developing plans for managing challenging behaviours. Students with autism do not learn social skills incidentally by observation and participation. It is generally necessary to target

specific skills for explicit instruction and to provide support for using the skills in social situations.

Additional information is provided in Chapter 4, *Strategies for Social Skills Training*.

Unusual behaviour

People with autism often demonstrate unusual and distinctive behaviours, including:

- restricted range of interests, and a preoccupation with one specific interest or object
- inflexible adherence to a non-functional routine
- stereotypic and repetitive motor mannerisms, such as hand flapping, finger flicking, rocking, spinning, walking on tiptoes, spinning objects
- preoccupation with parts of objects
- fascination with movement, such as the spinning of a fan, or turning wheels on toys
- insistence on sameness and resistance to change
- unusual responses to sensory stimuli

Implications for instruction

Many of the odd and stereotypical behaviours associated with autism may be caused by other factors, such as a hyper-sensitivity or hypo-sensitivity to sensory stimulation, difficulties in understanding social situations, difficulties with changes in routine, and anxiety. The instructional plan needs to incorporate strategies for expanding the student's interests, developing skills, understanding the student's responses to sensory stimuli, and preparing the student for planned changes.

In planning instruction, teachers need to consider the problematic behaviour and its function for that particular child—for example, gaining attention or avoiding something. It may not be possible to eliminate all repetitive behaviours. Successful teaching strategies for supporting students with autism often focus on making environmental adaptations to decrease an inappropriate behaviour, and/or helping the student to learn another more appropriate behaviour that will serve the same function.

Additional information is provided in Chapter 5, *Managing Challenging Behaviours*.

Learning

People with autism have a psycho-educational profile that is different from normally developing individuals. Studies show that there may be deficits in many cognitive functions, yet not all are affected. In addition, there may be deficits in complex abilities, yet the simpler abilities in the same area may be intact. Current research identifies the following cognitive features associated with autism:

- deficits in paying attention to relevant cues and information, and in attending to multiple cues
- receptive and expressive language impairments, particularly the use of language to express abstract concepts
- deficits in concept formation and abstract reasoning
- impairment in social cognition, including deficits in the capacity to share attention and emotion with others, and to understand the feelings of others
- inability to plan, organize, and solve problems



Minshew, N. J., *Autism as a Selective Disorder of Complex Information Processing*, 1998.

Some students with autism have stronger abilities in the areas of rote memory and visual-spatial tasks than they have in other areas. They may actually excel at visual-spatial tasks, such as putting puzzles together, and perform well at spatial, perceptual, and matching tasks. Some may be able to recall simple information, but have difficulty recalling more complex information.

Strength in visual-spatial skills has been described in personal accounts of individuals with autism. Temple Grandin suggests that some people with autism can more easily learn and remember information that is presented in a visual format, and that they may have problems learning about things that cannot be thought about in pictures. She explains that she has a visual image for everything she hears and reads, and that she “thinks in pictures.”

"When I was a child and a teenager, I thought everybody thought in pictures. I had no idea that my thought processes were different."

Temple Grandin, *Thinking in Pictures and Other Reports From My Life With Autism*, 1995, p.20

Students with autism may have difficulty comprehending oral and written information—for example, following directions or understanding what they read. Yet some higher-functioning individuals may be relatively capable of identifying words, applying phonetic skills, and knowing word meanings.

Some students may demonstrate strength in certain aspects of speech and language, such as sound production (phonology), vocabulary, and simple grammatical structures (syntax), yet have significant difficulty carrying on a conversation and using speech for social and interactive purposes (pragmatics).

A student who is high-functioning may perform numerical computations relatively easily, but be unable to solve mathematical problems.

Implications for instruction

These cognitive variations result in patterns of strengths and weaknesses in a student's academic performance, social interaction, and behaviour. Development of cognitive skills is usually uneven. The education program planned for the student should therefore be based on the unique combination of strengths and needs of that individual.

The professional literature on autism has documented deficits in attention and language development, problems in concept formation, and difficulties with memory for complex information. These characteristics, considered in combination with personal accounts of how individuals with autism are more visually oriented, suggest that visual material should be incorporated when teaching individuals with autism.

Suggestions for instructional strategies are provided in Chapter 4, *Teaching the Student with Autism*.

Unusual patterns of attention

People with autism often demonstrate unusual patterns of attention. Students can have a range of difficulties in this area. These have major implications for effective communication, social development, and attainment of academic skills.

Students with autism often have difficulty paying attention to relevant cues or information in their environment and may focus their attention only on a restricted part of the environment, to the exclusion of what is relevant. For example, a student may look at the ball but not at the person to whom the ball is to be thrown. Or a child may notice the insignificant details such as the staple in the corner of a paper, but not the information on the paper. This is referred to as *stimulus over-selectivity*. Another feature of autism is an impairment in the capacity to share attention between two things or people, which is referred to as *joint attention*.

Students may also have difficulty disengaging and shifting attention from one stimulus to the next, which may contribute to the characteristic rigidity and resistance to change. They may also demonstrate a short attention span.

Implications for instruction



Rosenblatt, J., et al.,
"Overselective responding: Description, implications, and intervention,"
in *Teaching Children With Autism: Strategies for Initiating Positive Interactions and Improving Learning Opportunities*, 1995.

Difficulties with attending may significantly influence the student's ability to develop effective social behaviour and language. For example, students with autism may respond to irrelevant social cues that have caught their attention, or they may attend to limited portions of a conversation and not understand the intent of what is being communicated. They may not attend to multiple cues in speech and language and so miss the important subtleties of the message.

Information presented to students with autism and instructional activities planned for them should be provided in a format that is clear, focusses their attention, and emphasizes the most relevant information. Individualized strategies for focussing the student's attention can be developed as part of the instructional plan. (See Chapter 4, *Teaching Students with Autism*, for suggestions.) Parents can provide valuable information when they share their methods of helping their children focus on things they need to learn. Ideally, the plan will include helping the student to manage these strategies themselves eventually.

Unusual responses to sensory stimuli

Awareness of different sensory experiences is central to understanding behaviours and planning programs for students with autism.

Students with autism usually differ from others in their sensory experiences. Responses to sensory stimulation may range from hyposensitivity to hypersensitivity. In some cases, one or more of the person's senses is either under-reactive (hyporeactive) or over-reactive (hyper-reactive). Environmental stimuli may be disturbing or even painful to someone with autism. This may apply to any or all types of sensory input. Personal accounts of autism have emphasized this.

The other characteristics associated with autism may be caused, in part, by a disorder in sensory processing. The extent to which sensory problems may contribute to other characteristics associated with autism is not certain. However, there is sufficient information to suggest that consideration be given to both the type and amount of sensory stimulation in the environment, and the individual's reaction to it.

Tactile system

The tactile system includes the skin and the brain. Information can be gathered by the skin through touch, temperature, and pressure. This information is interpreted by the person as pain, neutral information, or pleasure. The tactile system allows us to perceive and respond appropriately to our environment and have the appropriate reaction for survival. We pull away from something that is too hot and might harm us. We respond with pleasure to the warmth and pressure of a hug.

When students with autism are affected in the tactile system, they may withdraw when touched. This is called *tactile defensive*. They may overreact to the texture of objects, clothing, or food. The inappropriate response is the result of the person's tactile misperception, which can lead to behavioural problems, irritability, or withdrawal and isolation. Although some sources of stimulation may cause avoidance, other types and/or amounts of stimulation may have a calming effect.

Auditory system

Students with autism may be hyposensitive or hypersensitive to sounds. Parents and teachers report that seemingly innocuous sounds can cause extreme responses in some children with autism. This can be particularly problematic in a school setting, which normally includes so many different sounds. The scraping of a chair, bells between classes, intercom announcements, and sounds of machinery fill a normal school day. People with autism report that such sounds seem excruciatingly intense to them.

Visual and olfactory systems

Different responses to sensory stimuli may also be apparent in a student's reaction to visual information and smells. Some students may react to odours such as perfumes and deodorants. Others may use smell to seek out information about the surroundings in ways that we do not ordinarily expect.

Some students with autism cover their eyes to avoid the effect of certain lighting, or in response to reflections or shiny objects, while others seek out shiny things and look at them for extended periods of time.

Vestibular and proprioceptive systems

The inner ear contains structures that detect movement and changes in position. This is how people can tell that their heads are upright, even with closed eyes. Students with autism may have differences in this



Indiana Resource Center
for Autism, *Autism
Training Sourcebook*,
1997.

*"As I walk down the
street, I know what
everyone is having for
dinner by the smells com-
ing from the houses."*

—A 14-year-old boy with Asperger's
syndrome

orienting system so that they are fearful of movement and have trouble orienting themselves on stairs or ramps. They may seem strangely fearful or clumsy. The opposite is also true. Students may actively seek intense movement that upsets the vestibular system, such as whirling, spinning, or other movements that others cannot tolerate.

Through information derived from muscles and other body parts, people automatically know how to move or adjust positions efficiently and smoothly. Students who have problems integrating the body's information have odd posture and may appear clumsy or sloppy.

Implications for instruction

These unpleasant or aversive sensory experiences may contribute to some of the inappropriate behaviours that individuals with autism display. For example, people with severe sensory processing problems may entirely shut down to avoid aversive stimuli or over-stimulation. Tantrums may be related to the desire to escape situations that are over-stimulating. Self-stimulating behaviours may help the individual calm down when stimuli become overwhelming, by generating a self-controlled, repetitive stimulus.

Being aware of different experiences of sensory stimulation and integration is an important part of understanding behaviours of students with autism and planning programs for them. Teachers and families can work together to assess sensory responses and implement strategies to enhance them when the student is hyporesponsive, and to calm them when the student is hyper-responsive.



For more information on sensory integration, see Yack, E., et al., *Building Bridges Through Sensory Integration*, 1998.

Anxiety

Anxiety is not identified in the *DSM-IV* criteria. However, many people with autism, as well as their parents and teachers, identify anxiety as a characteristic associated with autism. This anxiety may be related to a variety of sources, including:

- inability to express oneself
- difficulties with processing sensory information
- fearing some sources of sensory stimulation
- high need for predictability, and having difficulty with change
- difficulty understanding social expectations

- fearing situations because they are not understood

Implications for instruction

Programs for students with autism often need to address anxiety, and what seems to contribute to it. Changes and adaptations can be made within the environment to reduce anxiety-arousing situations, and a variety of strategies can be used to help the individual to manage anxiety and cope with difficult situations.

See Chapters 4, 5, and 6 for suggestions.



For more information on autism and anxiety, see Harrington, K., *Autism: For parents and professionals*, 1998.

TEACHING STUDENTS WITH AUTISM

Chapter Four

No single method for teaching students with autism is successful for all students. Also, students' needs change over time, making it necessary for teachers to try other approaches. This chapter contains information about important areas of instruction and instructional approaches that have proved successful for teachers working with students with autism, including suggestions from the literature on autism. The chapter is divided into five sections:

- Instructional approaches
- Strategies for classroom management
- Strategies for communication development
- Strategies for teaching social skills
- Teaching functional skills

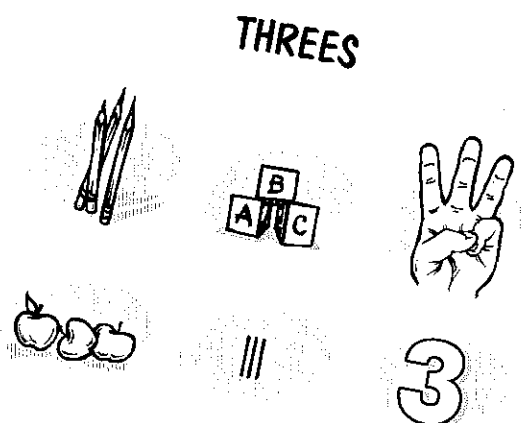
Instructional approaches

Visual approaches

The most strongly recommended approach for teaching students with autism is to use visual aids. Students often demonstrate relative strengths in concrete thinking, rote memory, and understanding of visual-spatial relationships, and difficulties in abstract thinking, social cognition, communication, and attention. Pictographic and written cues can often help the student to learn, communicate, and develop self-control.

One of the advantages of using visual aids is that students can use them for as long as they need to process the information. In contrast, oral information is transient: once said, the message is no longer available. Oral information may pose problems for students who have difficulty processing language, and who require extra time. In addition, it may be difficult for the student with autism to attend to relevant information and to block out background stimulation. Using visual supports enables the individual to focus on the message.

Visual aids and symbols range in complexity from simple and concrete to abstract. The continuum moves from real object or situation, to





Quill, K. A. "Visually cued instructions for children with autism and pervasive developmental disorders," *Focus on Autistic Behaviour*, 10(3), 1995.

Hogdon, L. A., *Visual Strategies for Improving Communication, Volume 1: Practical Supports for School and Home*, 1995.

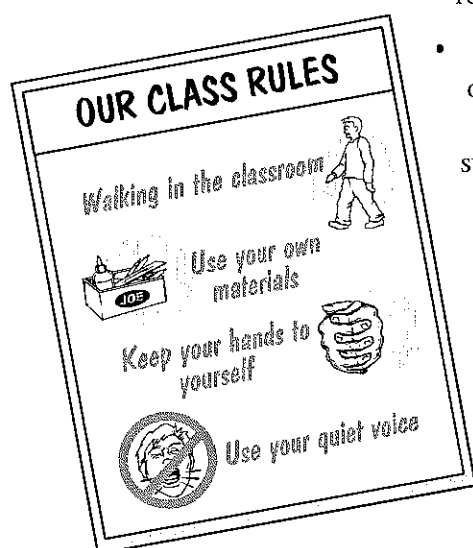
facsimile, colour photograph, colour picture, black and white picture, line drawing, and finally to graphic symbol and written language. Objects are the most simple, concrete form of aid. Graphic symbols, although far along the continuum in terms of complexity and abstraction, have been widely successful with students with autism. Software packages that provide quick access to graphic symbols and the ability to create customized symbols are available.

Visual supports can be used in a variety of ways in the classroom. However, to be successful, they must fit the student's level of comprehension by being at the appropriate point on the continuum of complexity. Using a line drawing to support learning when the student needs colour photographs in order to comprehend will only frustrate everyone.

Taking this caution into account, visual supports are very useful and can be employed to:

- organize the student's activity—daily schedules, mini-schedules, activity checklists, calendars, choice boards
- provide directions or instructions for the student—visual display of classroom assignments, file cards with directions for specific tasks and activities, pictographs and written instructions for learning new information
- assist the student in understanding the organization of the environment—labelling of objects, containers, signs, lists, charts, and messages
- support appropriate behaviour—posted rules and representations to signal steps of routines
- teach social skills—pictorial representations of social stories depicting a social situation with the social cues and appropriate responses, developed for a specific situation for the individual student (for further information on social stories, see the section in this chapter on strategies for teaching social skills)
- teach self-control—pictographs, which provide a cue for behaviour expectations

The key question to ask when planning an activity or giving an instruction is: how can this information be presented in a simple visual format? Choose visual aids on the basis of an understanding of the student and her or his abilities and responses.



Other approaches

Provide precise, positive praise while the student is learning

Give students precise information about what they do right or well; for example, “great colouring,” or “good finishing of that math problem.” Generalized praise may result in unintended learning that is hard to reverse. Students with autism may learn on one trial, so directing the praise to the very specific behaviour is important: “Sal, you are doing very well at multiplying these numbers.” Superstitious learning can occur if students mistakenly connect something they are doing with the praise. Saying “Sal, you are doing very well” when Sal is also swinging his feet while he does the math assignment might connect the feet swinging with the general praise.

Use meaningful reinforcements

Reinforcers can be anything from praise to tangible objects that increase the behaviour the student is to learn. Students with autism may not be motivated by common reinforcers that work with other students. They might prefer some time spent alone, time to talk to a preferred staff member, a trip to the cafeteria, an exercise routine (such as going for a walk), time to play with a desired object, music, playing in water, getting to perform a favourite routine, items that provide specific sensory stimulation, or sitting at the window.

It is important to know what works as reinforcement for each child. A preference profile that identifies the activities or other reinforcers that are preferred by the student can be helpful. This “likes and dislikes” list can be developed with the help of the family and shared with all service providers. See the appendix for a Likes and Dislikes chart that can be used to record a student’s preferences.

Plan tasks at an appropriate level of difficulty

Students with autism may be particularly vulnerable to anxiety and intolerant of feelings of frustration if they cannot perform the tasks assigned. Increasing the level of difficulty gradually and scaffolding or supporting learning (particularly with visual information rather than solely oral explanations) will assist in minimizing the student’s frustration.

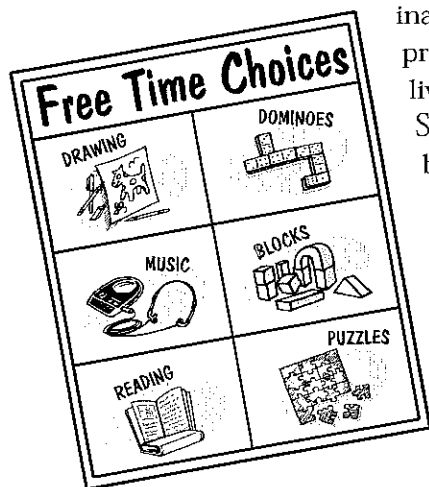
Use age-appropriate materials

It is important to honour the dignity of students with autism through the choice of instructional materials. Even if the instruction must be modified significantly, the learning materials should be appropriate to the age of the student.

Provide opportunities for choice

Because students with autism may be frequently frustrated by their inability to make themselves understood, they need instruction and practice in making good choices for themselves. Many parts of their lives may necessarily be highly structured and controlled by adults. Sometimes students continue to choose one activity or object because they do not know how to choose another.

Acceptable methods of providing choice for students who have limited ability to communicate need to be developed on an individual basis. Direct teaching of making choices may be helpful. Choice should be limited to one or two preferred activities until the student grasps the concept of choice. Open-ended choices will not enhance the student's skill at making choices, and may only frustrate him or her.

***Break down oral instructions into small steps***

When providing instruction for students with autism, teachers should avoid long strings of verbal information. As discussed above, supporting oral instruction with visual cues and representations will help students to understand.

Pay attention to processing and pacing issues

Students with autism may need longer to respond than other students. This may be linked to cognitive and/or motor difficulties. Students with autism may need to process each discrete piece of the message or request, and therefore need extra time to respond. Providing extra time generally, and allowing for ample time between giving instructions and student responses are both important tactics for supporting students with autism.

Use concrete examples and hand-on activities

Teach abstract ideas and conceptual thinking using specific examples, and vary the examples so that the concept is not accidentally learned as applying in only one way.

Use task analysis

Teachers and parents may need to break complex tasks down into sub-tasks and reinforce in small, teachable steps. For each step of a complex task, the student needs to have the requisite skills. These sub-skills may need to be taught and reinforced in sequence. For example, when teaching a self-help skill such as brushing teeth, the task may need to be broken down into sub-skills: getting the toothbrush and toothpaste, turning on the water, wetting the toothbrush, unscrewing the lid of the toothpaste, putting the toothpaste on the toothbrush, etc. Life skills, social skills, and academic skills can all be analysed and approached as tasks and sub-tasks, with each step taught and then linked to the next in a chain sub-tasks. See the appendix for an example of an instruction plan using task analysis.



Seip, J., *Teaching the Autistic and Developmentally Delayed: A Guide for Staff Training and Development*, 1996.

Use discrete trial methods

Using prompts to help students learn is an important element of instruction for some students with autism. Prompts may be physical, gestural, or verbal. They should only be used as long as they are needed, as students can become dependent on prompts. When using the discrete trial strategy, the instructor presents the stimulus for the desired behaviour (gives the directions or instructions), and prompts the student; the student responds, and then the instructor provides consequences based on behavioural principles. The prompt is often designed to model the desired behaviour or assist the student in performing it. Here is an example of a discrete trial format:

Behaviour objective: Identifying numbers, given an oral direction: Jackie will touch the card representing the correct number when presented with cards with the numbers 1, 2, 3, 4, and 5 on them.

STIMULUS	INSTRUCTOR PROMPT	STUDENT RESPONSE	CONSEQUENCE
Instructor says, "Touch 5."	Instructor taps the card with #5 on it.	Student touches the card with #5 on it.	Instructor smiles and says, "Good work, Jackie."
Instructor says, "Touch 5."	Instructor points at card with #5 on it.	Student touches card with #3 on it.	Instructor turns head and makes no verbal response.
Instructor says, "Touch 5."	Instructor taps the card with #5 on it.	Student flaps hands.	Instructor turns head and makes no verbal response.
Instructor says, "Touch 5."	Instructor points at card with #5 on it.	Student touches the card with #5 on it.	Instructor smiles and says "Good work, Jackie."

Introduce unfamiliar tasks in a familiar environment when possible

When it is not possible to introduce unfamiliar tasks in a familiar environment, prepare the individual for the new task and environment using aids such as pictures, videotapes, and/or social stories.

Organize teaching materials and situation to highlight what is important

Use organization aids and visual supports to:

- help the student attend to pertinent information, and
- teach new tasks

For example, remove extraneous materials from the desk or table before attempting to teach a skill. Or present only the text you want

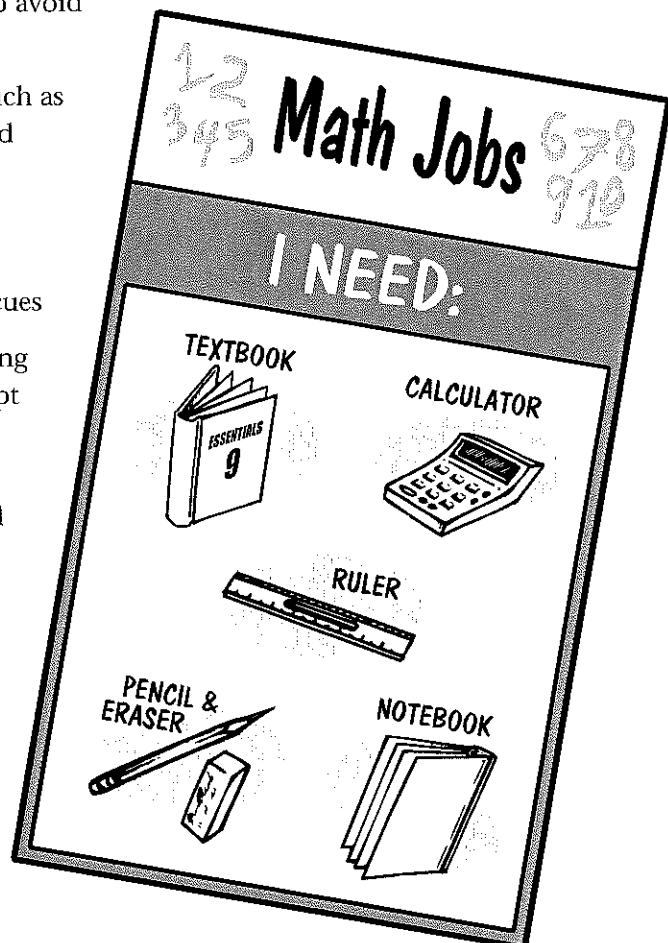
read rather than the whole book. Highlight the key words, such as character names in the text, so they are noticed.

Encourage independent effort and incorporate proactive measures to reduce the likelihood of becoming dependent on prompts

When students with autism are constantly supported, they may never develop the capacity to act independently. Since independence is a desired goal for all students, instruction should include strategies to decrease the need for adult prompting. Strategies include:

- using visual aids to decrease reliance on physical and verbal prompts from the parent, teacher, or teacher assistant
- planning ways to fade prompts
- ensure that the adult is not always positioned close to the student and that the same adults are not always present; positioning the adult away from the student and changing teacher assistants may help to avoid dependency
- providing visual organizational aids, such as schedules, task outlines, check lists, and charts, and involving the student in developing and using them, if feasible
- providing instruction to increase the student's awareness of environmental cues
- teaching in the environments containing the cues and reinforcement that prompt and maintain the behaviour

See the appendix for a chart explaining the hierarchy of prompts from high to low in level of intrusiveness.



Direct and broaden fixations into useful activities

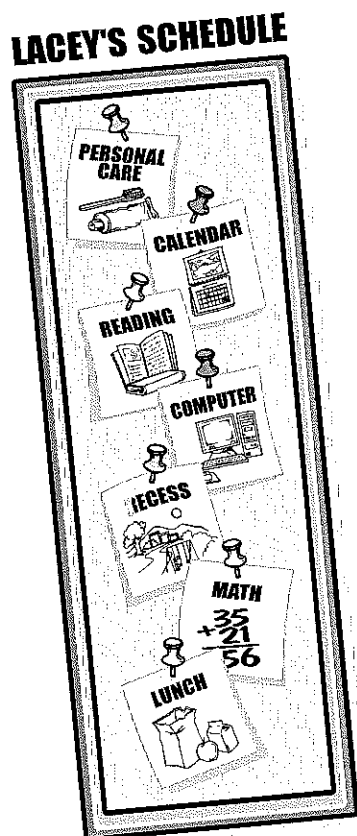
If the student is fixated on an object or a topic, such as a colour or shape, use it to teach a concept. A whole week's learning activities in writing and math can be centred on one topic—this is creative theme-based learning activities taken to the extreme.

Know the individual, and maintain a list of strengths and interests

Family members can provide valuable information for teachers about what students know and do at home or in the community. These interests and skills can be built upon both for instruction and for reinforcing successful learning and behaviour.

Develop talent and interest areas

If the student demonstrates a particular interest and strength in a specific area (e.g., music, drama, art, graphics, computer), provide opportunities to develop further expertise in that area. This may not only provide enjoyment and success, but may also lead to the development of skills for future employment.

**Strategies for classroom management*****Provide a structured, predictable classroom environment***

This is not to be confused with an authoritarian approach. The environment should be structured in order to provide consistency and clarity, so that students know where things belong and what is expected of them in a specific situation, and can anticipate what comes next.

Provide a customized visual daily schedule

The individualized schedule for a student with autism should fit comfortably into the overall classroom schedule. Vary tasks to prevent boredom, and alternate activities to reduce anxiety and possibly prevent some inappropriate behaviours. For example, alternate familiar, successful experiences with less preferred activities. It may be helpful to alternate large group activities with opportunities for calming down in a quiet environment. In addition, incorporating physical activity and exercise at points throughout the day is helpful. All planned activities can be charted in a visual form and posted at or near the desks of students with autism so that they can understand

changes in activities and know what to expect. The students can be helped to learn to use the schedule independently and staff can direct the student to the schedule when it is time to change activities to smooth transition times.

COMMON DIFFICULTIES WITH SENSORY SYSTEMS: OBSERVABLE BEHAVIOURS	
HYPER-REACTIVE BEHAVIOUR INDICATORS	HYP0-REACTIVE BEHAVIOUR
AUDITORY SYSTEM	
• easily distracted by background sounds • over-reacts to sounds • unpredictable reactions to sounds • holds hands over ears to block noise • screams or cries at sounds in the environment • responds physically as if sound is a threat • does not respond to any kind of sound	• does not respond to name being spoken • seems oblivious to sounds of surrounding activities • creates constant sounds as if to stimulate self • unsafe because does not react to sounds indicating potential danger
VISUAL SYSTEM	
• disturbed by bright lighting • avoids sunlight • follows any movement in the room with eyes • blocks field of vision with eyes • covers part of visual field—puts hands over part of the page in a book • responds physically to appearance of certain objects or colours	• unaware of the presence of other people • unable to locate desired objects, people • loses sight of people or objects when they move • cannot distinguish figure-ground relationships
TACTILE SYSTEM	
• touch defensive—does not like to be touched • avoids tasks with strong tactile element (clay, water play, paint, food preparation) • complains about discomfort of clothing • refuses to wear certain items—tugs at clothes • responds negatively to textures in foods, toys, furniture	• does not seem to grasp concept of personal space • does not seem to notice touch of others • frequently puts things into mouth • does not adjust clothing, which would seem to be an irritant • high pain threshold, unaware of danger because of low response to pain
VESTIBULAR SYSTEM	
• over-reacts to movement activities • has difficulties navigating on different surfaces (carpets, grass, etc.) • walks close to wall, clings to supports such as banisters • seems to be fearful when movement is expected, muscles seem tense • rigid about positioning of body, keeps head in same rigid angle • seems to become physically disoriented easily	• seems to need constant movement • rocks, travels in circles • seems to tire easily when engaged in movement activities • generally slow to move, lethargic in movement • takes long time to respond to directions to move

COMMON DIFFICULTIES WITH SENSORY SYSTEMS: OBSERVABLE BEHAVIOURS	
HYPER-REACTIVE BEHAVIOUR INDICATORS	HYPO-REACTIVE BEHAVIOUR
GUSTATORY & OLFACTORY SYSTEMS	
• eats a limited variety of foods • gags, refuses foods • difficulties with oral hygiene • spits out foods, medications • over-reacts to smells in environment • smell-defensive—will avoid places or people with strong odours	• seems to be constantly wanting food • licks objects in the environment • chews on objects inappropriately • high threshold for bad tastes—dangerous substances are not avoided • sniffs objects and people in unusual ways • does not seem to notice smells others notice

An inventory of the possible sensory factors can be used to help minimize the negative effect that sensory information may be having on students with autism. Parents and others who have experience with the student will be a valuable source of information about sensory difficulties. Here are some questions to ask and other points to consider when developing an inventory:

- Auditory:
 - Are there fans, loudspeakers, fire alarms, several people talking at once, air conditioners, bells, dogs barking, or scraping?
 - What is the general sound level and the predictability and repetitiveness of sounds?
 - What can be done to minimize the negative effect these stimuli may have on the student with autism in the class?
 - Consider the individual's comprehension of verbal information and the time typically required to process auditory information and shift attention between auditory stimuli.
- Visual:
 - Are there distracters, such as light, movement, reflection, or background patterns, that affect the student's ability to attend to the learning activity?
 - Consider the eye level of the student, the position of the teacher in relation to the student, and distracters that may interfere with attention.
 - Also consider the time required to shift visual attention.

"Autism cuts me off from my own body, so I feel nothing. It can also make me so aware of what I feel that it is painful."

—Donna Williams, an adult with autism, in *Somebody Somewhere*, 1994

- Careful attention to aversive visual stimuli and attempts to reduce the effect of these stimuli will assist managing the student's behaviour and help the student learn.
- Tactile:
 - Are there textures that seem to be aversive?
 - Are temperatures appropriate to minimize negative effect on the student?
 - Does the student demonstrate a need to explore through touch, and yet avoid being touched?
 - What is the level of ability or defensiveness in the use of certain objects intended to support instruction?
- Vestibular:
 - Consider the student's need to move and exercise.
 - What are the individual's reactions to movement?
 - How can the student's program incorporate needed movement without unduly jeopardizing the attention and learning of other students in the class?
- Gustatory and olfactory:
 - Consider the preferences in taste and smell of foods and other materials.
 - Decisions about activities should include consideration of the student's responses to the smell of materials.
 - Teaching the appropriate behaviour for snack or mealtimes will be affected by these preferences.

Note aspects of the tasks and activities that create frustration

Examine the instructional plan and non-instructional activities for problem areas that may result in sensory overload or frustration for the student. Make available sensory experiences that are calming for the student to accompany potentially frustrating tasks. Whenever possible, adapt tasks and materials to promote successful participation. When feasible, decrease environmental distracters and reduce activities that confuse, disorient, or upset the student and interfere with learning.



For more information on assessing sensorimotor systems of students with autism, see Dunn, W., "The Sensorimotor Systems: A Framework for Assessment and Intervention" in *Educating Children with Multiple Disabilities: A Transdisciplinary Approach* (1991) and Janzen, J., "Effects of the Autism Learning Style: Reception and Processing Problems" in *Understanding the Nature of Autism: A Practical Guide*, 1996.

Provide relaxation opportunities and areas

It may be necessary to have a calm, quiet, designated area where the student can go to relax. Relaxing for some students with autism may mean engaging in repetitive behaviours that have a calming affect on them. In some cases, students who crave certain repetitive movement, such as rocking or other self-stimulating movements, can be provided with a time and space where this movement is permitted.

Provide opportunities for meaningful contact with peers who have appropriate social behaviour

It will be necessary to teach appropriate social behaviour and to provide the student with situation-specific expectations for behaviour. More information on the development of social skills is provided in the “Strategies for developing social skills” section later in this chapter.

Opportunities for contact with peers may include:

- involving the student in shared learning arrangements
- pairing the student with buddies for walking down the hall, on the playground, and during other unstructured times
- varying peer buddies across time and activities, to prevent dependence on one child
- involving peers in providing individualized instruction
- arranging cross-age peer supports/buddies by assigning an older student to assist the student with autism
- pairing students while attending special school events such as assemblies and clubs
- facilitating involvement in after-school or extracurricular activities

If your school has an arrangement in which a class of older students is paired with a younger class, ensure that the older student with autism is also paired, and provide the necessary supports for success.

Plan for transitions and prepare the student for change

Students with autism often find changes in activity, setting, or planned routine very stressful. Visual schedules can be used to help them understand and co-operate with necessary changes. Social stories with illustrations can also be used to prepare the student for new situations.

For more information on classroom management, see Chapter 5, *Managing Challenging Behaviour*.

Tad's Week						
SUN	MON	TUES	WED	THURS	FRI	SAT
2	3	4	5	6	7	8
		MOVIE NIGHT	MOVIE NIGHT		PICNIC	

Strategies for communication development

Expanding the communication skills of students with autism is one of the greatest challenges for teachers and families. Most people are unaware of the complexity of normal communication, because children develop these skills automatically, usually by the age of three or four. Many students with autism have not developed the skills they need for spontaneous communication, and must therefore be taught. Helping students with autism develop communication skills—so that they can express their wants and needs, interact socially, share information, express emotions, and protest or escape aversive situations—is a priority.

Programs to facilitate the development of communication may begin in structured settings; however, promoting generalization and facility in using language requires that interventions take place in natural settings. Functional language skills are best taught in the social context where they will be used and where they have real meaning. The classroom and school environments provide a wealth of opportunities for developing functional communication within social contexts, and promoting generalization. However, opportunity alone will not address the communication needs of the student with autism. The specific skills requiring instruction and strategies for developing the targeted skills must be identified.

The school team, parents, and specialized professionals should collaborate to identify communication goals and objectives for the student with autism. The planned interventions should be based on the abilities and needs of the student. The speech and language pathologist can assist in assessment of communication skills and provide suggestions and strategies tailored to the unique needs and characteristics of the student.

Here are some general suggestions for assisting with communication:



For more information on developing communication, see Wetherby, A. M., and Prizant, B.M., (Eds.), *Autism Spectrum Disorders: A Developmental Transactional Perspective* (in press, 2000).

"Autism...stops me from finding and using my own words when I want to. Or makes me use all the words and silly things I do not want to say."

—Donna Williams, in *Somebody Somewhere*, 1994

- Focus on developing interaction and communication in the environments in which the child participates (e.g., classroom, playground, gym).
- Use sentences to talk to the student. Keep in mind that you are modelling speech as well as trying to communicate with the student.
- Use vocabulary appropriate to the student's comprehension capability. For students with more severe communication disability, choose familiar, specific, and concrete words, and repeat as necessary.
- Use language that is clear, simple, and concise. Figures of speech and irony or sarcasm will only confuse students with communication difficulties.
- Allow time for the student to process the information. It may be necessary to talk more slowly or to pause between words. The pace of speech depends on the ability of the individual student.

Learning to listen

Students with autism often need structured lessons on how to listen. Reinforcing listening efforts rather than assuming that listening is an expected and automatic behaviour may be necessary. Breaking listening down into components for the student and reinforcing each component may be helpful—for example, teaching the student to face the speaker, look at one spot (which does not mean they must make eye contact), and place hands in a planned position, and praising or otherwise rewarding each step.

Developing oral language comprehension

Use visual input to aid comprehension of oral speech. Visual aids may help obtain and maintain the student's attention. Accompanying spoken language with relevant objects, pictures, and other visual supports can help with comprehension. Experienced teachers of students with autism suggest the use of photographs to support understanding of the content of oral language communication. Interestingly, many students with autism use reading to support oral comprehension rather than the expected reverse of using oral language to support reading. This makes reading instruction even more significant for these students.

When working with students who are higher functioning, it is easy to assume that they understand information, particularly if they are able to repeat it. However, even though there may be good recall, the student may not grasp the intended meaning. It is important to check for comprehension.

Developing oral language expression

Students with autism may not develop traditional oral language, but most do develop some form of communication. It is important that people involved with the student have a thorough knowledge of the student's form of expression and that they adjust their expectations for communication accordingly. For students with limited oral expression, teachers and families should accept limited verbal attempts and non-verbal behaviour as communicative. A customized communication dictionary is a very useful tool in which staff and parents can document what the student says and what is meant, along with planned adult responses to language attempts. See the Appendix for a form that can be used to create a communication dictionary.

Even those students with autism who do have oral language may not add to their working oral vocabularies easily. Teachers and parents will need to teach new vocabulary in a variety of contexts and using a visually-based approach. Students need to be taught that:

- everything in our world has a name
- there are different ways of saying the same thing
- words can be meaningful in a variety of contexts, and
- learning to use words will help them communicate their needs and desires.

Students who rely on pictorial representations to communicate will need to learn that a drawing or representation has a name and that it can give direction, or tell us what to do. Understanding this is essential if visual systems are going to provide meaningful communication.

The student's education program should include situations that encourage different types of expression, such as:

- requests (e.g., for food, toys, or help)
- negation (e.g., refusing food or a toy, protesting when asked to do something, or indicating when the student wants to stop)

- commenting (e.g., labelling pictures in books, or objects from a box, greetings or play activities)

Developing conversation skills



Hunt, P., et al.,
"Acquisition of conversation skills and the reduction of inappropriate social interaction behaviours" in *Journal of the Association of Persons with Severe Handicaps*, 13, 1988.

Virtually all people with autism have difficulty with the pragmatics of communication—the interpretation and use of language in social situations. Even those individuals who have a good vocabulary and appear to have a command of the language may have a restricted understanding of social and conversational interactions.

For some students it may be necessary to provide structured teaching to develop the oral language needed for social and communicative play. This can be done by providing structured play opportunities that incorporate the student's interests. Modelling, physical prompts, visual cues, and reinforcement can be used to facilitate attention, imitation, communication, and interaction. To facilitate social communication, structure interactions around the student's activity preferences and routine. Encourage informal and formal communicative social exchanges during the day.

Simple drawings are an effective strategy for teaching conversation skills. These drawings illustrate what people say and do and emphasize what they may be thinking. A set of symbolic drawings can be used to represent basic conversational concepts, such as listening, interrupting, loud and quiet words, talk and thoughts. Colours may be incorporated to represent the emotional context. Pictures with scripts can also be used to develop conversation skills and communication appropriate to specific social contexts and situations.



Gray, C., *Comic Strip Conversations*, 1994.

People with autism have difficulty understanding subtle social messages and rules, and also have problems interpreting the non-verbal communication of others. It may be helpful to provide the student with a concrete rule when one does exist, and to present it in a visual format, by writing it down or incorporating it into a social story or comic strip conversation.

Students also need opportunities for social interactions and community-based experiences in order to practise the skills.

Echolalia

Some children with autism demonstrate *echolalia*, the literal repetition of words or phrases from language of other people. Young children use echolalia as part of normal language development. However, in autism, some learners seem to stop developing at this level of language growth.

Echolalia can be both immediate or delayed; that is, the student can repeat what was just heard or can repeat it later, sometimes many months or years later.

Immediate echolalia can be used as a teaching tool. The echolalic speech phrase can be shaped by using speech rules and by using the echolalic skill to model more appropriate language. For example, when a student echoes back questions, the teacher can shape the response by modelling the appropriate response and reinforcing the use of the appropriate response when the student echoes it. This type of strategy is highly individualized, and it may be appropriate to consult with the speech and language pathologist for specific suggestions for the individual student.

Delayed echolalic utterances may have no obvious meaning for the listener. Students with autism frequently repeat television commercials word for word. To understand the function of the language behaviour, it is helpful to think of it as a chunk of language that has been stored without regard for meaning. A situation or emotion may trigger the use of the speech, even if it seems to have no connection to the situation. It is important not to assume that the student understands the content of the echolalic speech being used. When possible, try to determine the situation that has elicited the speech and prompt the appropriate language to use for that situation. For example, when a student echoed the script from a TV coke advertisement, this meant that the student was thirsty. The teacher tested this possibility by verbal prompting with a question such as "You feel thirsty and want a drink?" Sometimes families and teachers never figure out a logical connection for delayed echolalic utterances.

Using alternative or augmentative communication systems

Many children can benefit from the use of an augmentative communication system. An augmentative communication system is any approach that supports, enhances, or adds to the way a person tells you something. It may be used with non-verbal students and also for students who have verbal expression, but appear unable to use speech in a functional way to express wants and needs. Augmentative communication systems can range from low-tech (those not requiring any power source such as electricity) to high-tech (those systems that require power).



Prizant, B., and Duncan, J., "The functions of immediate echolalia in autistic children," *Journal of Speech and Hearing Disorders*, 46, 1981.



Indiana Resource Centre for Autism, *Autism Training Sourcebook*, 1997.



For additional information on PECS, see Frost, L. A., and Bondy, A. S., *The Picture Exchange Communication System: Training Manual*, 1994.



Janzen, J. E., *Understanding the Nature of Autism: A Practical Guide*, 1996.

Alternative communication may include:

- direct movement of person or object to communicate (e.g., pulling a teacher to the door when the student wants to go outside)
- using gestures or body actions to convey meaning (e.g., shaking the head to express negativity)
- using real objects to convey messages (e.g., bringing a jacket to ask to go home)
- using picture representations (e.g., the Picture Exchange System, or PECS)
- using the voice without conventional words (e.g., saying “Ah-ah-ah” to indicate need for the toilet)
- using written messages by pointing at already written ones, or by writing (e.g., using a word processor to communicate)
- using sign language gestures from a conventional, non-verbal, formal language (e.g., American Sign Language or Signed English)

Deciding whether to implement an alternative or augmentative communication system, and selecting the type of system, are both decisions that should be made carefully and based on an assessment of the learner’s level of cognitive ability, skills, interests, and motor abilities. The spectrum of choices can be illustrated as a continuum:

highly abstract	signing
↓	written sentences
↓	written phrases
↓	written words
↓	drawings
↓	photographs
↓	gestures
↓	miniature objects
concrete	full-sized objects

Parents are key players in such decisions, as the communication system should be used both at school and at home to be effective. The school district speech and language pathologist (SLP) or another professional in the area of autism and augmentative communication systems

may be important sources of expertise to help with the decision. The teacher's role is often implementing the decision and supporting the student in learning to use it to supplement oral speech or as a substitute for speaking.

Ongoing research is showing promising results using computer technology as a means for communication and computer-assisted learning as a strategy for teaching communication skills.

Strategies for teaching social skills

Most students with autism would like to be part of the social world around them. They have a need to interact socially and be involved with others. However, one of the defining characteristics of autism is impairment in social interactions and social skills. Students with autism have not automatically learned the rules of interaction with others, and they are unable to follow these unwritten rules of social behaviour.

Many people with autism are operating on false perceptions that are rigid or overly literal. Recognizing these false perceptions can be very helpful in understanding the behaviour and needs of these students in social situations. The misperceptions include:

- rules apply in only a single situation
- everything someone says must be true
- when you do not know what to do, do nothing.

Imagine how overly literal misconceptions could seriously limit social interaction. It is a mistake to assume that students with autism understand any situation or a social expectation. They may be using an ineffective method of interacting because they do not know another more appropriate one, or they may be unable to distinguish between situations in order to select an appropriate behaviour.

Social skill development is an essential curricular area for students with autism, as well as a crucial component of any intervention plan for changing problem behaviours. In order to help students, it is necessary to carefully assess their social competencies to determine which social skills must be directly taught.

To develop social skills, students need to have the opportunity to participate and interact in a variety of natural environments where



Janzen, J. E.,
*Understanding the
Nature of Autism: A
Practical Guide*, 1996.

appropriate models, natural cues and stimuli, and functional reinforcers are available. Placement within integrated environments provides this access to peer models and social opportunities.

However, access to models and opportunities to develop social skills is not usually enough. In general, people with autism need explicit teaching to develop social skills and understanding of social situations. There are a variety of promising practices for supporting students with autism in developing social skills.

Using social stories

One of the most helpful methods for teaching social skills is the use of social stories, a strategy developed by Carol Gray. A social story is a description of a social situation that includes the social cues and appropriate responses, and that is written for a specific situation for the individual student. The story can be used for a variety of purposes, including:

- facilitating the inclusion of students in regular education classes
- introducing changes and new routines
- explaining reasons for the behaviour of others
- teaching situation-specific social skills
- assisting in teaching new academic skills.

Social stories can be created by parents, teachers, and other service providers. They are useful with students who have a level of cognitive functioning that allows them to understand the story. Non-readers can listen to social stories on cassette tapes. To be effective, a social story should describe a situation from the perspective of the student, direct the student to do the appropriate behaviour, and be in the voice of the student (i.e., from the “I” perspective).

The process begins with identifying student needs through observation and assessment. Once a difficult situation is identified, the author observes the situation and tries to understand the perspective of the student in terms of what will be seen, heard, and felt. The author then writes the story at an appropriate comprehension level and from the perspective of the student, and includes descriptive, directive, and perspective statements. (Descriptive sentences provide information on the setting, activity, and people involved; directive statements are positive statements about the desired response for a given situation; and

perspective statements provide a description of the possible reactions of others.)

The following formula provides a good rule of thumb for the content of a social story:

$$\begin{array}{rcl} 2 - 5 & \text{descriptive statements} & \\ + \underline{1} & \text{directive statement} & \\ = & \text{social story} & \end{array}$$

The most effective format for a story is a booklet with one or two sentences on each page, and a single page containing one main concept.

My Turn on the Computer

- p.1 If I wait for my turn on the computer, the other kids like me better.
- p.2 Everyone likes to have a turn on the computer.
- p.3 **When other kids are using the computer, I will be quiet and wait for my turn.**
- p.4 When I am finished on the computer, other kids can use it. That is okay, because I know I can use it the next day.
- p.5 When I wait for my turn on the computer, everyone will be happy.

There are three basic approaches for implementing a social story:

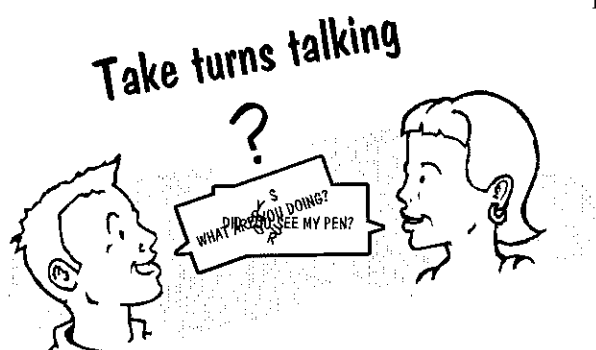
- For a student who reads independently, the story is read twice by an adult, followed by the student reading it back. Then the student reads it daily.
- If the student does not read, the story may be recorded on a cassette tape with a signal (i.e., bell) to turn the pages. The student is taught to “read” the story, and reads it daily. Symbols, drawings, or photographs can be included in the story to support meaning for the student.
- To incorporate modelling, the story can be videotaped. The story is read aloud on a videotape, with one page on the screen at a time.



Gray, C., *The Social Story Book*, 1993.

Teaching key social rules

Developing an understanding of the basic rules associated with a given situation will help the child to adapt to the social context, and may prevent increased anxiety and reduce the reliance on inappropriate coping behaviours. Critical social skills for which students with autism will likely need some type of direct instruction include:



- **Waiting**—Visual cues such as an object, pictures, and written words can provide concrete information to make waiting less abstract and more specific to the situation.
- **Taking turns**—This can be taught through the use of social stories as well as a picture or pictograph to cue the child. It may also be necessary to provide some instruction and rehearsal in turn-taking activities.
 - **Transitions**—Using social stories and providing warnings with visual cues, such as symbols that are understood by the student, can help the student make the transition from one activity to another. Transitions can be particularly difficult if the student has not completed the activity; the student may need to be prepared for the possibility of having to finish later.
- **Changing the topic in conversation**—Some students may stay on one topic and appear unable or unwilling to talk about anything else. Staying with one behaviour or topic in this way is referred to as perseveration. Visual rules, established time limits, and setting a time and place to engage in a favourite topic may help in teaching students when they need to end or change the topic.
- **Finishing**—It may help to teach students to use environmental cues, such as observing and following the behaviour other children. It may also be necessary to use a timer, and a method for checking their own work.
- **Initiating**—Social stories combined with photographs or pictures can be particularly useful for teaching a student how to approach others, ask for something, get into a game, say hello, and leave a situation if upset.

- Being flexible—Visual systems can be used to explain changes in a concrete way. If sequenced schedules or picture routines are used, a specific picture or symbol can be removed or crossed out, and another put in its place.
- Being quiet—Visual supports may be helpful in teaching the specific behaviours for being quiet, and teaching rules for specific situations.



Indiana Resource Centre
for Autism, *Autism
Training Sourcebook*,
1997.

Using Cognitive Picture Rehearsal

Another instructional strategy for teaching social skills that presents information in a visual format is Cognitive Picture Rehearsal. This method involves presenting a sequence of behaviours in the form of pictures or pictographs with an accompanying script. The student is guided through repeated practice of the sequence of behaviours.

Using peer support

Peers can assist students with autism in developing social skills. It may be helpful to educate the peers first, so that they better understand the behaviour of the autistic student. For example, the teacher may need to interpret the non-verbal communication, or explain that a specific activity is difficult for the student, and identify what peers can do to help. This can be done informally or in a more structured manner. Young children can be shown how to use specific prompts to initiate and maintain interaction with a classmate with autism. They may also need help communicating with the student. Peers should be reinforced for their role, just as the student with autism is reinforced for social interactions.

Peers can be helped to develop strategies to enhance the social competence of the child with autism. Pivotal Response Training (PRT) is one technique that has been used during recess breaks and has been successful in increasing interactions, initiation, varied toy play, and language use. PRT involves teaching typical peers to use strategies to:

- gain attention
- give choices to maintain motivation
- vary toys
- model social behaviour
- reinforce attempts



Groden, J., and
LeVasseur, P., "Cognitive
picture rehearsal: A sys-
tem to teach self-con-
trol," in *Teaching
Children with Autism:
Strategies to Enhance
Communication and
Socialization*, 1995.



For additional informa-
tion, see Quill, K.A.,
*Teaching Children with
Autism: Strategies to
Enhance Communication
and Socialization*, 1995.



Pierce, K., and Schreibman, L., "Using Peer Trainers to Promote Social Behaviour in Autism," *Focus on Autism and Other Developmental Disabilities*, 12(4), 1997.

- encourage conversation
- extend conversation
- take turns
- narrate play

Students can be provided with information on autism and tips for interacting with the student with autism. It is important that parents be involved in the decision to discuss autism with their child's peers. They may wish to preview any materials or may want to be involved in the presentation.

Using social skills training groups

Students with autism may also benefit from social skill instruction within a small-group structured format. There are a variety of social skills training programs and resources available.

Promising programs include an assessment that is used to identify skills for instruction. Lessons follow a similar format in each of the social skills curricula:

- identifying the skill and skill components, and when it is used
- modelling the skill
- role play
- opportunities to practise, and
- strategies for generalization

Although these curricula are not developed specifically for children with autism, they can be used in combination with appropriate adaptations and supports. In addition, there may need to be a particular emphasis on the strategies for facilitating generalization of targeted skills.

Integrated play groups

Integrated play groups can provide opportunities for younger students with autism to interact with their age peers, and create a natural environment for incidental teaching of social skills. Play groups provide natural situations in which children with autism use language to express wants, practise being near other children, and imitate social interactions between non-disabled peers.



For additional information on social skills training groups, see McGinnis, E. and Goldstein, A., the *Skillstreaming series*, 1997; Elliot, S., and Gresham, M., *Social Skills Intervention Guide*, 1991; and Walker, et al., *Walker Social Skills Curriculum: The Accepts Program*, 1988.

Teaching self-monitoring/managing skills

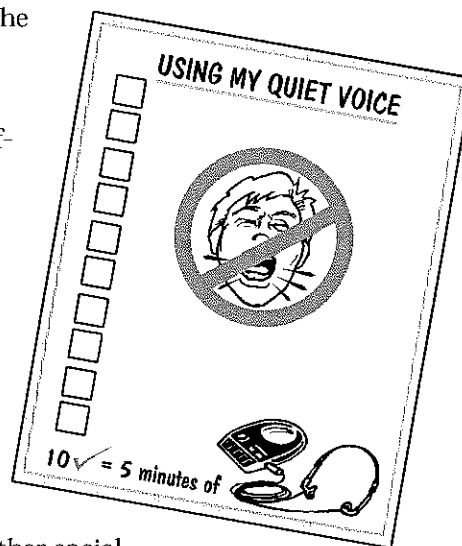
The ultimate goal for all students, including those with autism, is to increase independent participation in a variety of environments with effective social skills. One way to increase independence in higher-functioning students with autism is to teach self-management procedures, in which students monitor their own behaviour in order to earn positive reinforcement. Studies have shown that in the process of the student collecting his or her own self-monitoring data, the desired behaviour increases. The accuracy of the self-monitoring may not be as important as the process and awareness it builds in the student. The process for teaching self-management is as follows:

1. Define the target behaviour that the student will self-monitor.
2. Identify reinforcers that function successfully for the individual.
3. Create a self-monitoring method for the student to collect data (e.g., a chart, stickers, or some kind of low-tech counter device).
4. Teach the student the target behaviour and how to use the self-monitoring method to record the performance of the behaviour.
5. Increase the student's independence by gradually reducing adult intervention and having the student self-manage behaviour.

Supporting the development of friendships

Optimally, the aim of developing specific social skills is to enable the student to interact with others in a variety of settings, and to facilitate the development of social opportunities and relationships. Students who demonstrate basic social skills may still have difficulty establishing connections with other children and maintaining interactions with peers. Teachers and parents may facilitate further social interaction through:

- encouraging a friend to play with the child at home
- helping the student join school clubs with support as needed to participate
- teaching the child to observe other children to follow what to do
- encouraging co-operative games



- modelling how to relate to the child, and educating other students in the class
- encouraging prospective friendships
- providing enjoyment at break times
- doing projects and activities that illustrate the qualities of a good friend
- helping the student to understand emotions through direct teaching of how to read people's faces and body language and respond to cues that indicate different emotions

See Chapter 5, *Managing Challenging Behaviour*, for further suggestions on developing appropriate social behaviour.

Teaching functional skills

One of the fundamental goals of schooling is that students acquire the skills they need to function as independently as possible in the world. This may be even more important for students with such disabilities as autism, because they have significant difficulties in acquiring independent functioning skills.

For students who have needs in the area of functional skills development, goals for these skills should be identified in the student's education program and included in the IEP. The same instructional approaches and strategies used for other areas can be applied to instruction in functional skills.

In the field of special education, educators have developed a variety of models for the domains of functional skills. Although these models differ in some ways, they basically include five domains:

- domestic, or self-care
- functional academics
- vocational, or job skills
- social, including leisure skills
- community, including travel and using services

Schools and families should co-ordinate the planning of instruction for functional skills, so that instruction at both home and school is consistent and efficient. Some of these skills involve the most personal

areas of a person's life, so sensitivity and care need to be used in planning with parents or other caregivers.

Self-care

The same kinds of instructional strategies can be applied to instruction in the areas of self-care as with communication or social skills. Students with autism, particularly those who also have intellectual disabilities, often need direct instruction in personal hygiene, grooming, and dressing. Toileting can be an area requiring significant planning and instruction. Planning meals, food preparation, and even eating may be an appropriate part of a student's program. Household skills required for living independently (e.g., doing laundry, caring for clothing, and cleaning) may be taught or reinforced in the school program. Handling money and budgeting are essential skills for older students.

Functional academics

Being able to apply the basic academic skills of reading, writing, and mathematics to real life situations is another important area of functional skill development for many students with autism. Students need to learn how to communicate personal information such as their name, birth date, address, and telephone number. They need to recognize important signs and instructions in writing, such as labels and street signs. Using measurement for weight, volume, distance, and size; counting; using calendars; and telling time are all mathematical literacy skills that are of critical importance to independent functioning.

Vocational skills

Students with autism usually require instruction in basic skills needed for the world of work. These skills are broad and overlap with all the other areas. Independent adults need to have skills such as:

- being punctual and reliable in attendance at the work site
- following a job routine, and completing duties as assigned
- understanding task completion
- following safety procedures
- accepting direction and correction
- responding appropriately to persons in authority

- completing a cleanup routine
- dressing in appropriate work attire and using appropriate grooming
- using job site leisure time appropriately (lunch, breaks)

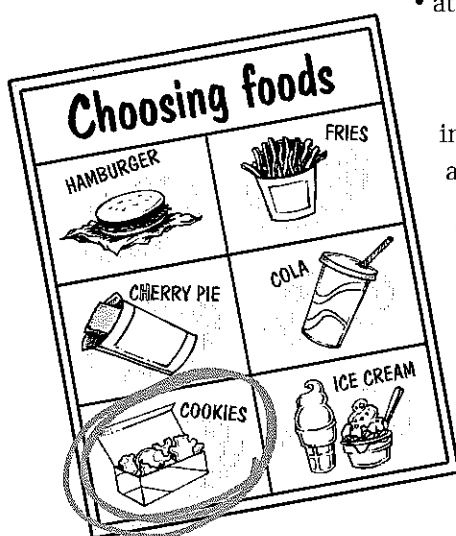
Leisure skills

Education programs for students with autism often include a recreational component, in recognition of the fact that they need help in developing a positive use of their spare time. For some individuals whose disabilities preclude employment in the future, leisure activities make up an even more significant part of their daily routines as adults.

Participation in leisure activities can vary from full participation to partial participation, depending on the needs of the individual. Finding ways to plan for meaningful partial participation is a challenge facing families and schools.

Leisure activities include:

- team sports (e.g., soccer)
- individual sports (e.g., bowling)
- arts activities (e.g., music)
- attending performances (e.g., theatre, movies)
- nature activities (e.g., camping, hiking)
- participating in organized groups (e.g., Scouts)
- attending social events (e.g., dances)



Developing activities that can be enjoyed at home is also important. Students may need support in finding and learning activities such as:

- using a television, stereo, and VCR
- caring for pets
- playing games such as cards
- sewing, knitting, or doing other crafts

Community skills

Safety is a major concern for many students with autism. It is important to consider safety issues in planning for them as they develop independence in the community. Social skills are of course closely connected to community skills. Possible areas for consideration in planning community skills instruction include:

- using public transportation
- finding community services such as pools, recreation centres, and banks
- managing pedestrian rules and understanding traffic
- using public facilities such as washrooms
- restaurant skills such as choosing and ordering food



For more information on planning instruction for community skills, see Gray, C., *What's Next: Preparing the Student with Autism or Other Developmental Disabilities for Success in the Community*, 1992.

SUMMARY OF SUGGESTIONS FOR TEACHING STUDENTS WITH AUTISM

INSTRUCTIONAL APPROACHES	
Use visual methods of teaching	• post daily schedules • create individualized mini-schedules • provide activity checklists • create choice boards • label objects and containers • post classroom rules, with illustrations or symbols • create visual representations of steps in personal or class routines • provide visual cues to support oral information or teacher directions
Additional teaching approaches	• break tasks into component parts and teach each component • carefully word praise so that it is behaviour-specific • provide reinforcers that produce desired student response • use discrete trial strategies • use behaviour-shaping strategies • plan tasks at appropriate level of difficulty • use age-appropriate materials • provide opportunities for choice • use simple oral instructions supported by visual information • pace tasks at student's level • allow adequate time for processing information • wait with patience for student response • employ student interests to motivate and teach new skills • highlight important information

STRATEGIES FOR CLASSROOM MANAGEMENT

- provide a structured, predictable classroom environment
- customize a schedule and locate it at student's desk
- adapt the classroom environment to eliminate, as much as possible, stimuli that elicit problem behaviour
- devise strategies to minimize the effect of stimuli that cannot be eliminated
- adapt tasks and materials to avoid student frustration
- provide a relaxation area
- provide opportunities for student to be in contact with peers who model class routines
- plan carefully for transitions between activities and locations

STRATEGIES FOR COMMUNICATION DEVELOPMENT

Learning to listen	• provide structured lessons in listening • break down listening into behaviour components and reinforce each component
Developing oral comprehension	• use visual aids (photographs, pictures, objects, etc.) • use gestures with oral communication • pair written language with oral communication
Developing oral expression	• reinforce attempts to communicate • provide structured instruction of new vocabulary supported with visual aids • help student understand that everything in the environment has a name • use desired objects or activities to encourage expression • provide classroom situations in which comments are elicited
Developing conversation skills	• model appropriate skills, and have other students model desired skill • provide opportunities for structured play interactions • use discussions of routines to practise skills • teach students the correspondence between behaviours and thoughts • encourage and reinforce informal conversation • use prepared scripts to teach social conversations • teach rules for social discourse

STRATEGIES FOR SOCIAL SKILLS DEVELOPMENT

- use social stories
- teach components of key social skills
- provide planned practice and reinforcement for skills
- provide practice of skills using picture cue-cognitive picture rehearsal
- use peer support
- use social skills training groups
- support the development of friendships

STUDENTS WITH ASPERGER'S SYNDROME

Chapter Six

Asperger's syndrome (AS) has some of the same characteristics as autism. Although AS shares some characteristics with higher-functioning autism, it also has some unique features, and a different developmental progression and prognosis.

In British Columbia, the designation for supplemental funding of students with a medical diagnosis of AS should be based on the nature and extent of the disabling condition and the educational interventions required. These students may be reported to the ministry in High Incidence categories such as "Moderate Behaviour Disorders" or "Severe Learning Disabilities," or, in a very few cases, in the "Severe Behaviour" category or a Low Incidence category.

According to *DSM-IV* criteria, the child diagnosed with AS must meet the criteria for social impairment, repetitive activities, and age of onset, but have normal cognition and early language development. AS involves fewer symptoms than autism and is considered to be a mild to moderate disability, while autism is usually considered to be a severe disability.

Learning and behavioural characteristics of students with Asperger's syndrome

AS is characterized by a qualitative impairment in social interaction. People with AS may be keen to relate to others, but do not have the skills, and may approach others in peculiar ways. They frequently lack understanding of social customs and may appear socially awkward, have difficulty with empathy, and misinterpret social cues. They often have the same difficulties as individuals with autism in understanding that other people have their own perceptions, thoughts, and feelings. People with AS are also poor incidental social learners and need explicit instruction in social skills.

Although children with AS usually speak fluently by the time they enter Kindergarten, they often have problems with the complexities of language, including:

- pragmatics (the use of language in social contexts)
- semantics (multiple meanings)
- prosody (the pitch, stress, and rhythm of speech)



Myles, B. S., and Simpson, R. L., *Asperger Syndrome: A Guide for Educators and Parents*, 1998.



For more information on reporting students with medical conditions, see the B.C. Ministry of Education document *Special Education Services: A Manual of Policies, Procedures and Guidelines*, Section E, 1995.



American Psychiatric Association, *Diagnostic and Statistical Manual for Mental Disorders* (4th Ed.), 1994.

Klin, A., and Volkmar, F. R., "Asperger's Syndrome," in *Handbook of Autism and Pervasive Developmental Disorders* (2nd Ed.), 1997.

One of the common characteristics of people with AS is that they have trouble carrying on social conversations. They may have an advanced vocabulary and talk incessantly about a favourite subject. The topic may be somewhat narrowly defined and the person may have difficulty switching to another topic.

People with AS may have problems communicating with others because they do not naturally learn the rules of conversation. They may:

- interrupt or talk over the speech of others
- make irrelevant comments
- have difficulty initiating and terminating conversations
- use speech characterized by a lack of variation in pitch, stress and rhythm
- use overly pedantic or formal speech, particularly as the student reaches adolescence
- stand too close when talking to someone
- stare, use abnormal body posture, or use other strange body language
- fail to understand gestures and facial expressions of others



T. Attwood, *Asperger's Syndrome: A Guide for Parents and Professionals*, 1998.

Students with AS are of average to above-average intelligence and may appear quite capable. Many students with AS are relatively proficient in their knowledge of facts, and may have extensive factual information about a subject that absorbs them. However, they demonstrate relative weaknesses in comprehension and abstract thought, as well as in social cognition. Consequently, they do experience some academic problems, particularly with:

- reading comprehension
- problem solving
- organizational skills
- concept development
- making inferences and judgements

In addition, they often have difficulty with cognitive flexibility—that is, their thinking tends to be rigid—and with adapting to change or failure. They do not readily learn from their mistakes.

Students with AS often have difficulties with co-ordination. An estimated 50–90% of people with AS have problems with motor co-ordination.

The affected areas may include locomotion, balance, manual dexterity, handwriting, rapid movements, rhythm, and imitation of movements.

Students with AS may share common characteristics with autism in terms of responses to sensory stimuli. They may be hypersensitive to some stimuli and may engage in unusual behaviours or repetitive behaviours to obtain a specific sensory stimulation.

Students with AS may also be inattentive, easily distracted, and anxious. Many students diagnosed with AS are inattentive and easily distracted and have received a diagnosis of Attention Deficit/ Hyperactivity Disorder at one point in their lives. Anxiety is also a characteristic associated with this syndrome. It may be difficult for the student to understand and adapt to the social demands of school. Appropriate instruction and support can help to alleviate some of the stress.

Strategies for teaching students with Asperger's syndrome

Some of the strategies for teaching students with autism may be applicable to students with AS. However, it is important to consider the unique learning characteristics of the individual student, provide support when needed, and build on the student's many strengths.

The following chart identifies some specific learning difficulties and suggests a number of possible classroom strategies, adapted from a variety of sources in the literature.



T. Attwood, *Asperger's Syndrome: A Guide for Parents and Professionals*, 1998.



Myles, B. S., and Simpson, R.L., *Asperger Syndrome: A Guide for Educators and Parents*, 1998.



For more information on Attention Deficit/Hyperactivity Disorder, see the Ministry of Education handbook *Teaching Students with Attention Deficit/Hyperactivity Disorder: A Resource Guide for Teachers*, 1996.

Learning Difficulty	Classroom Strategies
Difficulties with language - tendency to make irrelevant comments - tendency to interrupt - tendency to talk on one topic and to talk over the speech of others - difficulty understanding complex language, following directions, and understanding intent of words with multiple meanings	- use <i>Comic Strip Conversations</i> (Gray, 1994) to teach conversation skills related to specific problems - teach appropriate opening comments - teach student to seek assistance when confused - teach conversational skills in small group settings - teach rules and cues regarding turn-taking in conversation and when to reply, interrupt, or change the topic - use audio-taped and video-taped conversations - explain metaphors and words with double meanings - encourage the student to ask for an instruction to be repeated, simplified, or written down if he does not understand - pause between instructions and check for understanding - limit oral questions to a number the student can manage - watch videos to identify non-verbal expressions and their meanings
Insistence on sameness	- prepare the student for potential change, wherever possible - use pictures, schedules, and social stories to indicate impending changes
Impairment in social interaction - has difficulty understanding the rules of social interaction - may be naive - interprets literally what is said - difficulty reading the emotions of others - lacks tact - has problems with social distance - has difficulty understanding "unwritten rules" and once learned, may apply them rigidly - lacks awareness of personal space	- provide clear expectations and rules for behaviour - teach (explicitly) the rules of social conduct - teach the student how to interact through social stories, modelling and role-playing - educate peers about how to respond to the student's disability in social interaction - use other children as cues to indicate what to do - encourage co-operative games - provide supervision and support for the student at breaks and recess, as required - use a buddy system to assist the student during non-structured times - teach the student how to start, maintain, and end play - teach flexibility, co-operation, and sharing - teach the students how to monitor their own behaviour - structure social skills groups to provide opportunities for direct instruction on specific skills and to practise actual events - teach relaxation techniques and have a quiet place to go to relax - model and practise appropriate personal space
Restricted range of interests	- limit perseverative discussions and questions - set firm expectations for the classroom, but also provide opportunities for the student to pursue his own interests - incorporate and expand on interest in activities

Poor concentration • is often off task • is distractible • may be disorganized • has difficulty sustaining attention	• provide frequent teacher feedback and redirection • break down assignments • use visual organizers, semantic mapping, and outlining • provide timed work sessions • reduce homework assignments • seat at the front of the classroom • use non-verbal cues to get attention
Poor organizational skills	• use personal schedules and calendars • maintain lists of assignments • help the student use "to do" lists and checklists • place pictures on containers and locker • use picture cues in lockers
Poor motor co-ordination	• involve in fitness activities; student may prefer fitness activities to competitive sports • take slower writing speed into account when giving assignments (length often needs to be reduced) • provide extra time for tests • consider the use of a computer for written assignments, as students may be more skilled at using a keyboard
Academic difficulties • usually average to above-average intelligence • good recall of factual information • areas of difficulty include problem solving, comprehension, and abstract concepts • often strong in work recognition and may learn to read very early, but has difficulty with comprehension • may do well at math facts, but not problem solving	• do not assume that the student has understood simply because she or he can re-state the information • be as concrete as possible in presenting new concepts and abstract material • use activity-based learning where possible • use graphic organizers such as semantic maps, webs • break tasks down into smaller steps or present in another way • provide direct instruction as well as modelling • show examples of what is required • use outlines to help student take notes and organize and categorize information • avoid verbal overload • capitalize on strengths (e.g., memory) • do not assume that student has understood what he or she has read—check for comprehension, supplement instruction, and use visual supports
Emotional vulnerability • may have difficulties coping with the social and emotional demands of school • easily stressed because of inflexibility • prone to anxiety • often have low self-esteem • may have difficulty tolerating making mistakes • may be prone to depression	• may have rage reactions and temper outbursts • provide positive praise and tell the student what she or he does right or well • teach the student to ask for help • teach techniques for coping with difficult situations and for dealing with stress, such as relaxation strategies • use rehearsal strategies • provide experiences in which the person can make choices • help the student to understand her or his behaviours and reactions of others • educate other students • use peer supports such as buddy system and peer support network

Sensory sensitivities • most common sensitivities involve sound and touch, but may also include taste, light intensity, colours, and aromas • types of noise that may be perceived as extremely intense are: • sudden, unexpected noises such as a telephone ringing, or fire alarm • high-pitched continuous noise • confusing, complex, or multiple sounds, such as in shopping centres	• be aware that normal levels of auditory and visual input can be perceived by the student as too much or too little • keep the level of stimulation within the student's ability to cope • avoid sounds that are distressing, when possible • use music to camouflage certain sounds • minimize background noise • use ear plugs if noise or reaction is very extreme • teach and model relaxation strategies and use of diversions to reduce anxiety • provide opportunities and space for quiet time • arrange for independent work space that is free of sensory stimuli that bother the student
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